Autonomous Agents for Novel Architecture Discovery: Evaluating Expressivity, Efficiency, and Complexity

Recent progress in sequence model architectures has been driven by theoretical understanding of expressive power, and formal analyses have characterized which language classes different architectures can recognize (Merrill and Sabharwal, 2023; Merrill et al., 2024). In parallel, a different paradigm—AI coding agents—has shown increasing capability for long-horizon engineering tasks, and can be applied to research (Lu et al., 2024). These developments offer complementary strengths for testing new architectures: theory provides motivation and guarantees, while autonomous research provides scalability and breadth. It remains unclear whether the two can be meaningfully combined: *can an agentic approach systematically explore and advance expressive power in neural architectures?*

We hypothesize that agents may be well-suited to this problem. Improving neural architectures increasingly requires synthesizing ideas across disparate fields—numerical linear algebra, automata theory, GPU kernel optimization—that rarely appear together in human expertise. Agents can continuously monitor new publications, maintain a growing catalog of computational primitives, and systematically explore combinatorial pairings that human researchers might overlook. In addition, experiments in expressivity can often be validated with less compute relative to later stages of training. Perhaps most importantly, agents can run dozens of experiments in parallel, rapidly filtering a large hypothesis space to surface promising directions for human attention.

To test this hypothesis, we design a multi-agent system, and we propose two case studies to evaluate the system against, with results to be presented at the poster. Our system comprises four coordinating agents: a *trick search agent* that mines computational primitives from literature (matrix factorizations, parallelization schemes, algebraic structures); a *research agent* that synthesizes these into architectural proposals with hypotheses, mathematical formulations, and success criteria; an *experiment agent* that implements and runs minimum viable experiments (MVEs); and a *log agent* that analyzes outcomes and updates strategy. To date, the system has cataloged 381 tricks from 485 papers, generated 175 proposals, and implemented 117 MVEs spanning structured state transitions, input-dependent gating, memory compression, and kernel fusion strategies.

After a defined work period or token budget, we will assess the agent along these two case studies, to evaluate how well the system performed:

Case Study 1: Parameter-efficient associative recall. Multi-Query Associative Recall (MQAR; Arora et al., 2024) tests a model's ability to store and retrieve key-value pairs—a capability central to in-context learning. We ask: can agents discover architectures that achieve higher MQAR accuracy at fixed parameter count than published baselines (Gu and Dao, 2024; Yang et al., 2025)? *Quantitative*: We will plot accuracy vs. parameters for agent-discovered architectures against baselines, measuring whether any expand the Pareto frontier. *Qualitative*: We will analyze whether agents discover interpretable mechanisms (e.g., novel memory update rules, structured state transitions) or merely tune hyperparameters.

Case Study 2: State-tracking with formal guarantees. Tracking cumulative state through group composition (S_5, A_5) requires non-commutative dynamics that diagonal SSMs provably cannot express (Merrill et al., 2024). We ask: can agents discover architectures that solve state-tracking while maintaining sub-quadratic complexity? *Quantitative*: We will measure accuracy on S_5 composition vs. asymptotic complexity, comparing to the known hierarchy (diagonal fails, dense succeeds). *Qualitative*: We will examine whether agent proposals connect to formal results—e.g., do they rediscover that orthogonal or group-structured transitions are necessary, or find novel sufficient conditions?

Criterion	Target	Achieved	Status
Nyström copy accuracy	$> 90\%$	99.25%	✓ Pass
Full copy accuracy	$> 95\%$	99.08%	✓ Pass
Accuracy gap	$< 5\%$	-0.18%	✓ Pass
Memory compression	$O(mn) < O(n^2)$	$O(20) < O(64)$	✓ Pass
Layer 0 approx error	< 0.1	0.859	× Fail
Layer 1 approx error	< 0.1	0.907	× Fail

Table 1: Example result: Nyström landmark compression for chunkwise SSM (Experiment 025), inspired by Xiong et al. (2021). The agent hypothesized that Nyström compression could reduce inter-chunk state transfer cost while preserving accuracy. Four of six criteria passed—but the two failures revealed that the model *co-adapts* to compression rather than learning intrinsically low-rank transitions, contradicting the original assumption.

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